MIL-E-1/1338B(NAVY)
29 January 1973
SUPERSEDING
MIL-E-1/1338A(NAVY)
16 January 1961

MILITARY SPECIFICATION SHEET

ELECTRON TUBE, KLYSTRON

TYPE 6584

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Reflex, CW, tunable frequency range 5,100 to 5,900 MHz, power output 135 mW

ABSOLUTE RATINGS:

Parameter:	Ef	Ers	Er	Irs	Ehk
Unit:	V	Vdc	Vdc	mAdc	Vdc
Maximum:	6.8	330	-300	35	150
Minimum:	5.8	---	---	---	---

PHYSICAL CHARACTERISTICS:

Dimensions:	See figure 1		
Cathode:	Unipotential, oxide coated	Output coupling:	Coaxial lead
Base:	Modified, small wafer octal, 8 pin, B-21, low-loss phenolic	Cooling:	Convection
		Mounting position:	Any

TEST CONDITIONS:

Parameter:	Ef	Ers	Er	Irs	Ehk	F
Unit:	V	Vdc	Vdc	mAdc	Vdc	MHz
Test condition 1:	6.3	300	-85 to -235 (Note 2)	25	0	F2
Test condition 2:	6.3	300	-100	25	0	F2
Test condition 3:	6.3	300	-85 to -235 (Note 2)	25	0	F1
Test condition 4:	6.3	300	-85 to -235 (Note 2)	25	0	F3

FREQUENCY		
F	MHz	%
F1	5,100	±1
F2	5,500	±1
F3	5,900	±1

GENERAL:

Qualification - Required

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Qualification inspection							
1041	Shock	4,7	3	10 G; t = 32 ms	ΔF	---	4.0	MHz
4027	Temperature coefficient	6	3		ΔF/ΔT	---	+0.07 -0.08	MHz/°C
4302	Frequency change (warmup)		3	t1 = 20 seconds; t2 = 30 minutes	ΔF	---	15	MHz
	Quality conformance inspection, part 1	1	-					
---	Operation bump	5	1		ΔPo/Po	---	0.10	---
4250	Power output (1)	3	3		Po	70	---	mW
4213	Reflector voltage (1)	-	3		Er	-100	-160	Vdc
4213	Reflector voltage (2)	-	4		Er	-140	-210	Vdc
4280	Electronic tuning range (2)	3	4	Er = 50% Po	ΔF	28	---	MHz
4214	Cathode emission	-	1	Ef = 5.8 V	ΔIk/Ik	---	0.15	---
4229	Total reflector current	-	2		Ir	---	7	μAdc
4229	Reflector-leakage current	-	2		Ir	---	5	μAdc
4229	Reflector-gas current	-	2		Ir	---	2	μAdc
1256	Electrode current (resonator)	-	2	Nonoscillating	Irs	---	32	mAdc
4223	Mechanical tuning	-	1		F	5,100	5,900	MHz
	Quality conformance inspection, part 2							
1031	Operation vibration	4	3		ΔF	---	±1.5	MHz
1211	Insulation of electrodes	-	1	500 Vdc; tube cold	Rk-rs Rh-rs	2.0 2.0	--- ---	Meg Meg
1336	Heater-cathode leakage	-	3	Ehk = 45 Vdc	Ihk	---	100	μAdc
1301	Heater current	-	1		If	450	550	mA
4250	Power output (2)	3	4		Po	120	---	mW
4280	Electronic tuning range (1)	3	3	Er/50% Po	ΔF	34	---	MHz
4231	Electronic tuning hysteresis	-	1		---	---	0.05	%

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METHOD	REQUIREMENT OR TEST	NOTES	TEST	CONDITIONS	SYMBOL	LIMITS		UNIT
						MINIMUM	MAXIMUM	
	Quality conformance inspection, part 2 -Continued							
4215	Reflector voltage modulation coefficient	-	1	ΔEr 3 V	ΔF/ΔEr	0.6	1.2	MHz/V
---	Beam voltage modulation coefficient	-	1	ΔErs 20 V p to p ac	ΔF/ΔErs	---	0.320	MHz/V
	Quality conformance inspection, part 3							
---	Life test	-	1	Group D	t	500	---	hrs
---	Life-test end point:							
4250	Power output (2)	3	4		Po	100	---	mW

NOTES:

1. Unless otherwise specified, the AQL for all tests listed under quality conformance inspection, part 1, shall be 1.0, inspection level II.
2. Adjust Er/maximum Po.
3. The TUT shall be fixed firmly in a suitable socket by clamps which make contact at the flange adjacent to the base. The measurements on the tube in an oscillating state should be made with the output line coupled into a 50-ohm coaxial line having a VSWR of 1.15:1 or less, using an adapter per RT-1738, or equivalent.
4. The TUT shall be capable of meeting all quality conformance inspection, part 1 tests, after vibration and shock tests.
5. This test shall be made with a pendulum consisting of a steel ball 5/8-inch (15.88 mm) in diameter suspended by a wire 0.02-inch (0.51 mm) in diameter, with the distance between the point of suspension and the center of the ball 4-1/4 inches (107.95 mm), or with an equivalent pendulum. The ball shall be suspended so that when at rest it touches the tube at approximately one-half the distance between the extreme limits of the tube projecting above the socket. In this position, the point of suspension of the ball shall be directly above the center of the ball. The ball shall be permitted to swing freely through an arc of 60° in a plane through the axis of the tube and from such an angle that the ball strikes only the tube envelope.
6. The TUT shall be operated in accordance with the conditions specified in power output (1). The initial frequency shall be allowed to stabilize by operation at practically constant ambient temperature for at least 10 minutes. The temperature range for the test shall be over the range of 0° to 50°C in still air. The rate of change of ambient temperature shall not be greater than 2°C per minute. The temperature coefficient measurement shall consist of the average of the last two of three temperature runs as specified above.
7. ΔF to be defined as difference between frequency after final shock, and stabilized frequency prior to initial shock.

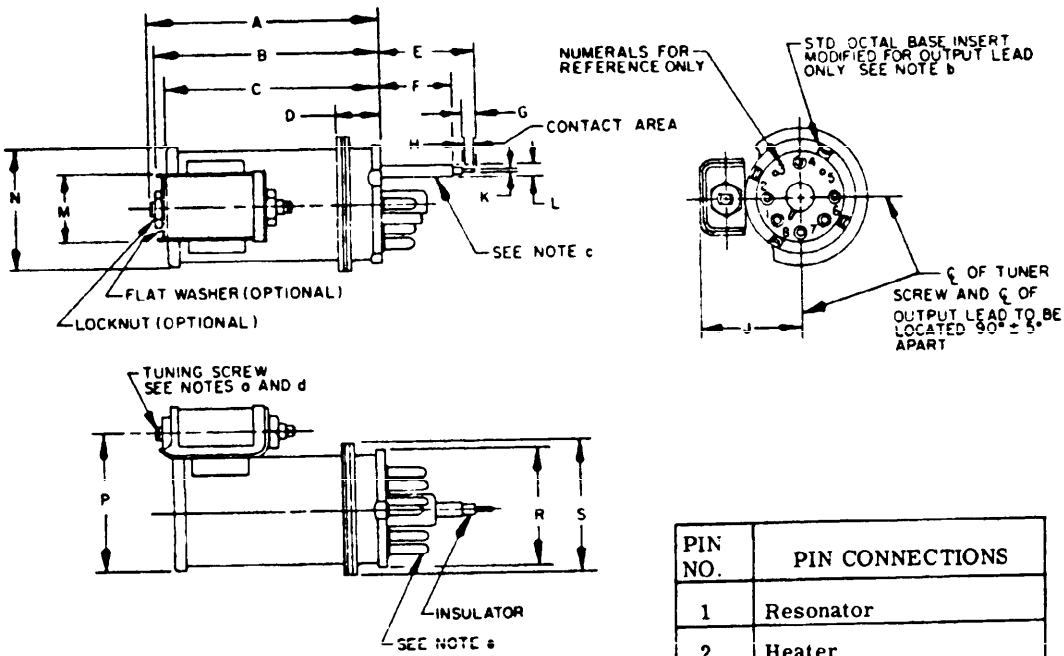
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PIN NO.	PIN CONNECTIONS
1	Resonator
2	Heater
3	
4	Output lead (coaxial)
5	
6	Reflector
7	Heater
8	Cathode

Tube: Shell resonator

FIGURE 1. Outline drawing of electron tube type 6584.

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses.	
	Minimum	Maximum
Quality conformance inspection, part 1		
A		2.984 (75.79)
D	.280 (7.11)	.344 (8.74)
F	.874 (22.20)	.984 (24.99)
G	.171 (4.34)	.201 (5.11)
M		.860 (21.84)
N		1.312 (33.32)
S	1.500 (38.10)	1.624 (41.25)
Quality conformance inspection, part 2		
B		2.687 (68.25)
C		2.500 (63.50)
E	1.203 (30.56)	1.233 (31.32)
J		1.313 (33.35)
K	.035 (.89)	.045 (1.14)
L	.135 (3.43)	.145 (3.68)
P	1.563 (39.70)	1.687 (42.85)
R	1.271 (32.28)	1.311 (33.30)
NOMINAL		
H	.156 (3.96)	

NOTES:
a. The tuning screw shall be lubricated with oildag or equal noncorrosive lub. It shall be capable of being operated smoothly thru entire range without binding.
b. This base shall be capable of being inserted in a gage same as RMAGES-1 except that No. 4 pin shall be .160 (4.06 mm) dia. to depth of 1.312 (33.32 mm) min.
c. Nickel (30 MSI silver plated permissible).
d. Maximum permissible torque of tuning screw 25 inch-ounces.
e. Solder on sides of base pins shall not be such that the pin and solder exceed the maximum dia of pin by more than .002 (.05 mm) and solder shall not extend more than .125 (3.18 mm) beyond the end of pin.

FIGURE 1. Outline drawing of electron tube type 6584 - Continued.

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